Meiosis

Exam code: 0610 & 0970

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Written by: Phil
Reviewed by: Lára Marie McIvor
Updated on 3 October 2024

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Meiosis: Extended

Help me understand

Extended Tier Only

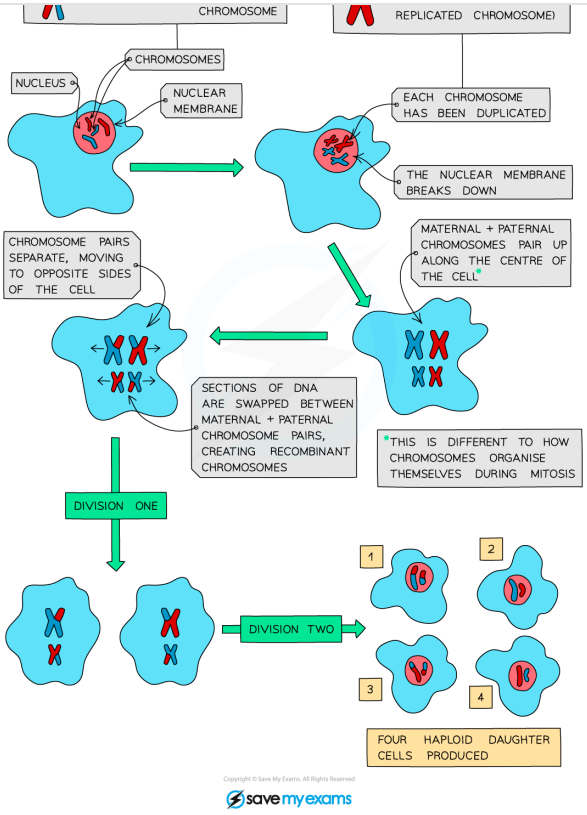
- **Meiosis** is a type of nuclear division that **gives rise to cells that are genetically different**
- It is used to produce the **gametes** (sex cells)

- The number of chromosomes must be **halved** when the **gametes** (sex cells) are formed
- Otherwise there would be double the number of chromosomes after they join at fertilisation in the zygote (fertilized egg)
- This halving occurs during **meiosis,** and so it is described as a **reduction division** in which the **chromosome number is halved from diploid to haploid**, resulting in **genetically different cells**
- It starts with chromosomes doubling themselves as in mitosis and lining up in the centre of the cell
- After this has happened the cells divide twice so that only one copy of each chromosome passes to each gamete
- We describe gametes as being **haploid** - having half the normal number of chromosomes
- Because of this double division, meiosis produces **four** haploid cells

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The process of cell division by meiosis to produce haploid gamete cells

Process:

- Each chromosome makes identical copies of itself (forming X-shaped chromosomes)
- First division: chromosomes pair up along the centre of the cell, recombination occurs and then cell fibres will pull the pairs apart, each new cell will have one of each recombinant chromosome pair
- Second division: chromosomes will line up along the centre of the cell, cell fibres will pull them apart (as with mitosis)
- A total of four haploid daughter cells will be produced

Importance:

- Production of gametes e.g. sperm cells and egg cells, pollen grains and ovum
- Increases genetic variation of offspring

- Meiosis produces **variation** by forming **new combinations** of maternal and paternal **chromosomes** every time a gamete is made, meaning that when gametes fuse randomly at fertilisation, each offspring will be different from any others

Differences between Mitosis & Meiosis

DAUGHTER CELLS ARE DIPLOID	DAUGHTER CELLS ARE HAPLOID
DAUGHTER CELLS ARE GENETICALLY IDENTICAL TO EACH OTHER AND TO THE PARENT CELL	DAUGHTER CELLS ARE GENETICALLY DIFFERENT FROM EACH OTHER AND THE PARENT CELL
ONE CELL DIVISION OCCURS	TWO CELL DIVISIONS OCCUR

Examiner Tips and Tricks

Questions on cell division often ask for differences between mitosis and meiosis. Learn two or three and remember to BE SPECIFIC when giving your answer.

You should also know the reasons for a specific type of cell division taking place and the types of cells where each happen.

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